TECHNICAL QUESTIONS

Question Pool • Easy, Moderate and Difficult

SET A — EASY (10 Questions × 5 Marks)

Q1. Output Prediction

Question: What is the output?

```
int a = 5;
int b = 2;
printf("%d", a * b + a);
```

> **Clue: Follow the order of arithmetic operations.**

Exact Answer: 15

Q2. C Programming

Question: What will be the output?

```
int x = 10;
if(x % 2 == 0)
printf("Even");
else
printf("Odd");
```

> **Clue: % gives the remainder.**

Exact Answer: Even

Q3. Number System

Convert the binary number **10110** into decimal.

> **Clue: Use powers of 2 from right to left.**

Exact Answer: 22

Q4. Loop

Question: How many times will **Hello** be printed?

```
for(int i = 1; i <= 10; i += 2)
{
printf("Hello ");
}
```

> **Clue: Write the values taken by i.**

Exact Answer: 5 times

Q5. Python

Question: What is the output?

```
x = 10
y = 3

print(x // y)
print(x % y)
```

Clue: // gives the quotient and % gives the remainder.

Exact Answer: 3 and 1

Q6. DBMS

Which key uniquely identifies each record in a table?
A) Foreign Key B) Primary Key C) Candidate Key D) Composite Key

Clue: Think of a unique student ID.

Exact Answer: B) Primary Key

Q7. Computer Fundamentals

Which component temporarily stores data and programs currently being used by the CPU?
A) Hard Disk B) RAM C) ROM D) Pen Drive

Clue: Its contents are lost when power is switched off.

Exact Answer: RAM

Q8. HTML

Which HTML tag is used to create a hyperlink?
A) <link> B) <a> C) <href> D) <url>

Clue: Think of the anchor element.

Exact Answer: <a>

Q9. Logic

Find the next number:
2, 6, 12, 20, 30, ?

Clue: Look at the differences.

Exact Answer: 42

Q10. C Programming

Question: What is the output?

```
int a = 10;
printf("%d", a++);
```

Clue: ++ is a post-increment operator.

Exact Answer: 10

SET B — MODERATE (10 Questions × 10 Marks)

Q11. Output Prediction

Question: What is the output?

```
int x = 1;
for(int i = 1; i <= 3; i++)
{
x = x * 2;
}
printf("%d", x);
```

Clue: The multiplication happens three times.

Exact Answer: 8

Q12. Array

Find the **second-largest distinct number**:
12, 5, 18, 18, 9, 15

Clue: Do not count 18 twice.

Exact Answer: 15

Q13. C Programming

Question: What is the output?

```
int a = 5;
int b = 10;

a = b;
b = a;

printf("%d %d", a, b);
```

Clue: Track the values after each assignment.

Exact Answer: 10 10

Q14. Python List

Question: What is the output?

```
a = [10, 20, 30, 40]
print(a[1] + a[-1])
```

Clue: -1 refers to the last element.

Exact Answer: 60

Q15. SQL

Question: Consider the table:

Name — Marks
Ravi — 75
Anu — 90
Kiran — 65
Meena — 90

What will this query return?

```
SELECT COUNT(*) FROM Student WHERE Marks >= 75;
```

Clue: Count the students satisfying the condition.

Exact Answer: 3

Q16. Algorithm

What is the main purpose of **binary search**?
A) Sort an array B) Search efficiently in a sorted array C) Reverse an array D) Find duplicate elements

Clue: Binary search repeatedly divides the search space.

Exact Answer: B

Q17. Number System

What is the decimal value of **1101■**?

Clue: 1×8 + 1×4 + 0×2 + 1×1.

Exact Answer: 13

Q18. Debugging

Question: Find the error:

```
int x = 10;
if(x > 5)
{
printf("Greater")
}
```

Clue: Look at the end of the printf statement.

Exact Answer: A semicolon (;) is missing.

Q19. Networking

Which device is primarily used to connect different networks and forward packets between them?
A) Switch B) Router C) Hub D) Repeater

Clue: Your home network uses one to connect to the Internet.

Exact Answer: Router

Q20. Logic + Programming

Question: What is the output?

```
x = 5
for i in range(1, 4):
x = x + i
print(x)
```

Clue: range(1,4) gives 1, 2, 3.

Exact Answer: 11

SET C — DIFFICULT (10 Questions × 15 Marks)

Q21. Array Problem

Given **[3, 4, -1, 1, 2, 6]**, find the **smallest positive integer that is missing**.

Clue: Start checking from 1.

Exact Answer: 5

Q22. C — Nested Loops

Question: What is the output?

```
int count = 0;
for(int i = 1; i <= 3; i++)
{
for(int j = 1; j <= i; j++)
{
count++;
}
}
printf("%d", count);
```

Clue: Count the inner-loop executions for each value of i.

Exact Answer: 6

Q23. SQL

Given employee salaries: **25000, 40000, 35000, 50000, 40000**. What is the **second-highest distinct salary**?

Clue: Ignore duplicate salaries.

Exact Answer: 40000

Q24. Recursion

Question: What is the output?

```
int fun(int n)
{
if(n == 0)
return 1;
return n * fun(n - 1);
}
printf("%d", fun(4));
```

Clue: Expand the function until n becomes 0.

Exact Answer: 24

Q25. Algorithm

An array contains **[2, 7, 11, 15, 3, 6]**. Find **all pairs whose sum is 9**.

Clue: Check which number each element needs to reach 9.

Exact Answer: (2,7) and (3,6)

Q26. C Output

Question: What is the output?

```
int x = 5;
if(x > 3)
{
if(x < 10)
printf("A");
else
printf("B");
}
else
{
printf("C");
}
```

Clue: Both conditions must be checked.

Exact Answer: A

Q27. DBMS

Which normal form removes **partial dependency**?
A) 1NF B) 2NF C) 3NF D) BCNF

Clue: Think about dependencies on part of a composite key.

Exact Answer: B) 2NF

Q28. Operating System

A program has CPU burst = **5 ms**, I/O burst = **10 ms**, CPU burst = **3 ms**. Ignoring context-switching time, what is the total CPU time required?

Clue: Add only the CPU bursts.

Exact Answer: 8 ms

Q29. Number + Logic

Find the next number:
1, 2, 6, 24, 120, ?

Clue: Each number is multiplied by an increasing integer.

Exact Answer: 720

Q30. Programming Logic — Challenge

Question: What is the output?

```
x = 0
for i in range(1, 6):
if i % 2 == 0:
```

```
x = x + i
else:
x = x - i
print(x)
```

Clue: Process 1, 2, 3, 4, 5 one at a time.

Exact Answer: -3